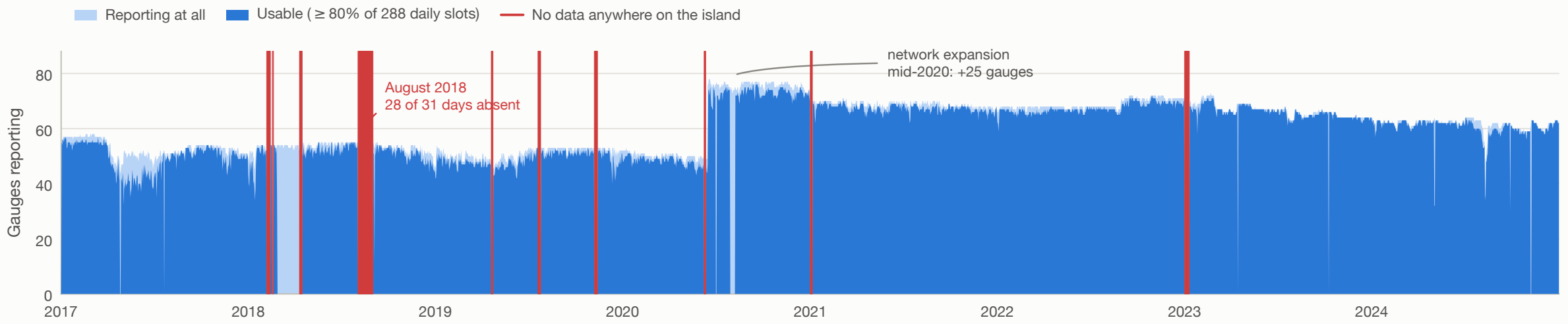
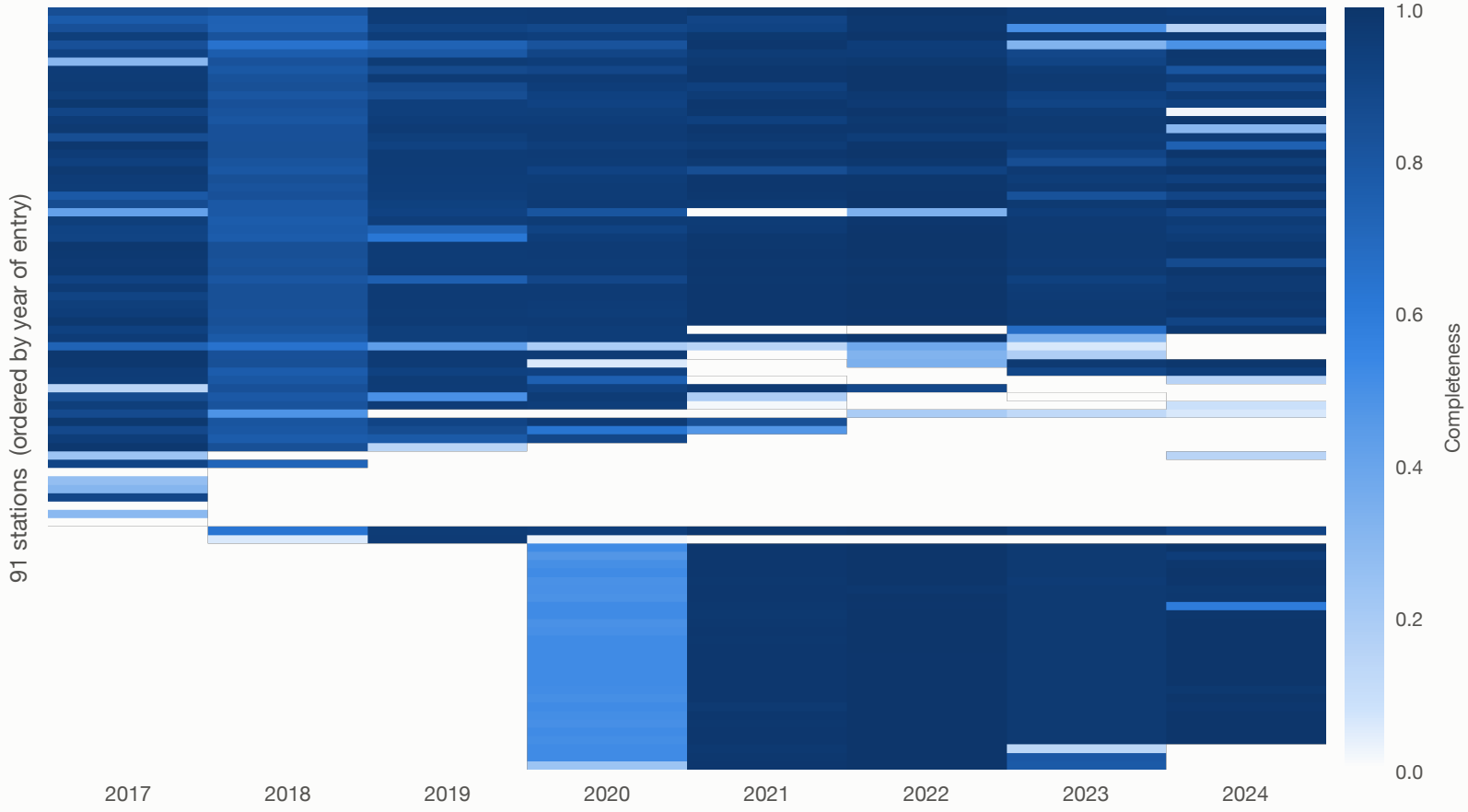
A changing gauge network, not a changing climate, is the first thing the raw data shows

Before any weather signal can be read, the record has to be audited. The network triples in places and vanishes in others, so island-wide figures in this report are areal means over gauges that pass a completeness rule, never raw averages over rows.

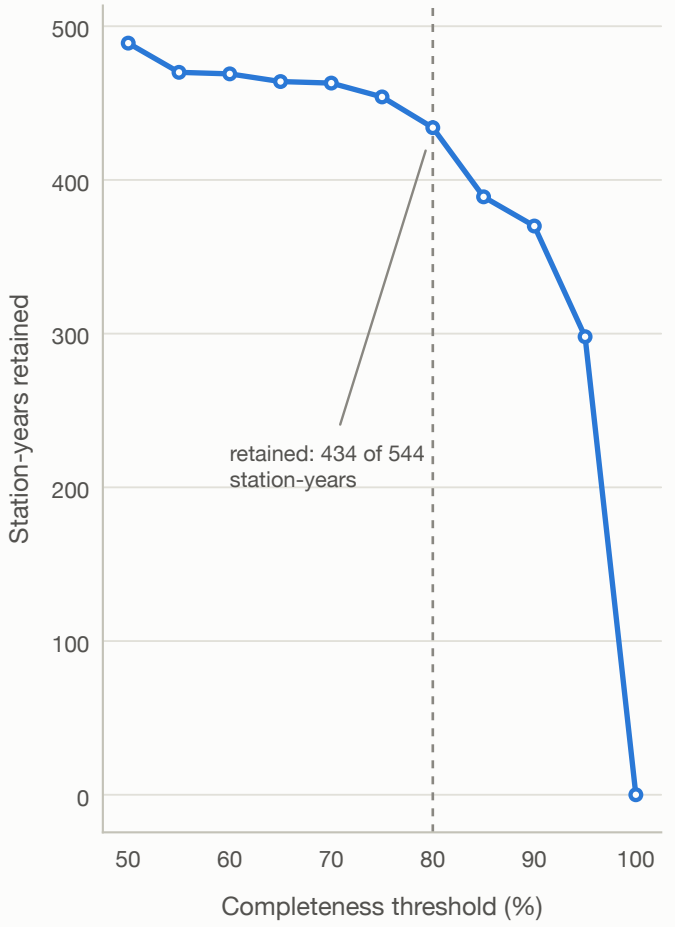
a The observing network is not constant: it grows, and it drops out



b Per-station annual completeness



c Cost of the QC rule

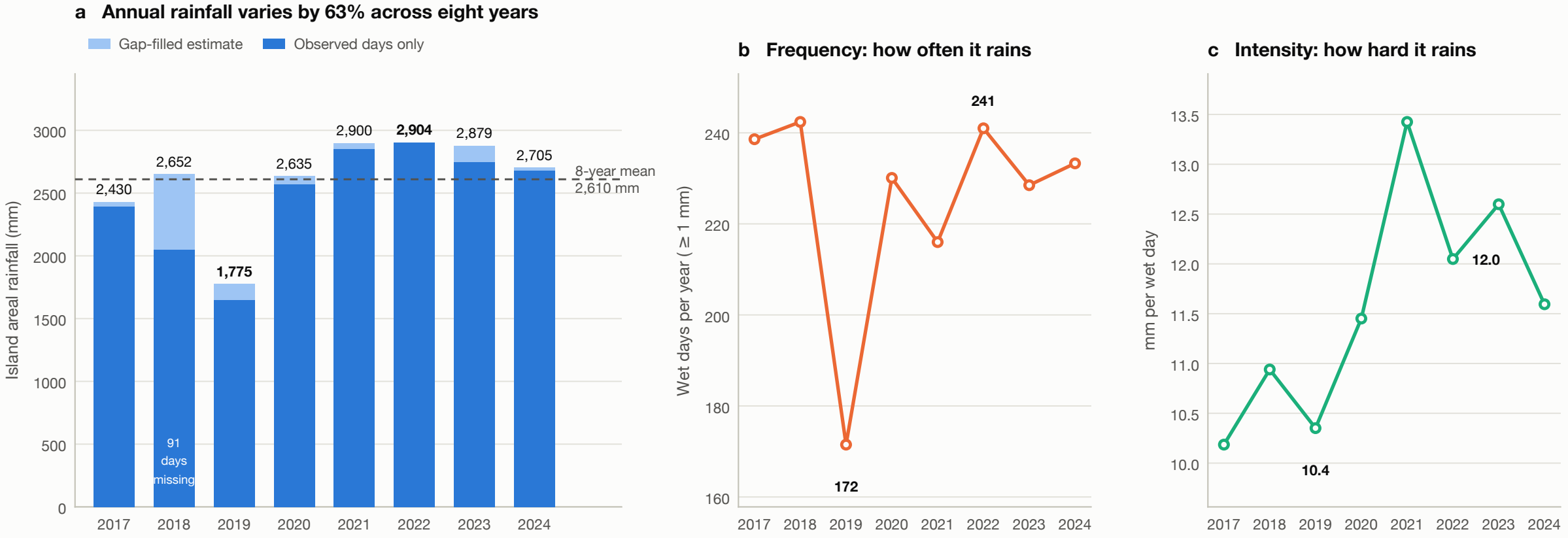


Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.

QC rule: a station-day needs ≥230 of 288 five-minute slots; an island-day needs ≥20 such stations.

2019 was dry because it rained less often, not because the rain got lighter

Splitting the annual total into frequency × intensity separates the two ways a year can be dry. The 2019 deficit (1,775 mm, a positive-IOD drought year) is a collapse in wet-day count, 172 days against a 225-day norm, while the rain that did fall was no lighter than 2017's. Panel c carries a separate signal worth noting: per-wet-day intensity climbs steadily after 2019, so the network is seeing fewer but heavier wet days in recent years. Panels b and c are drawn on their own axes rather than twinned onto panel a.



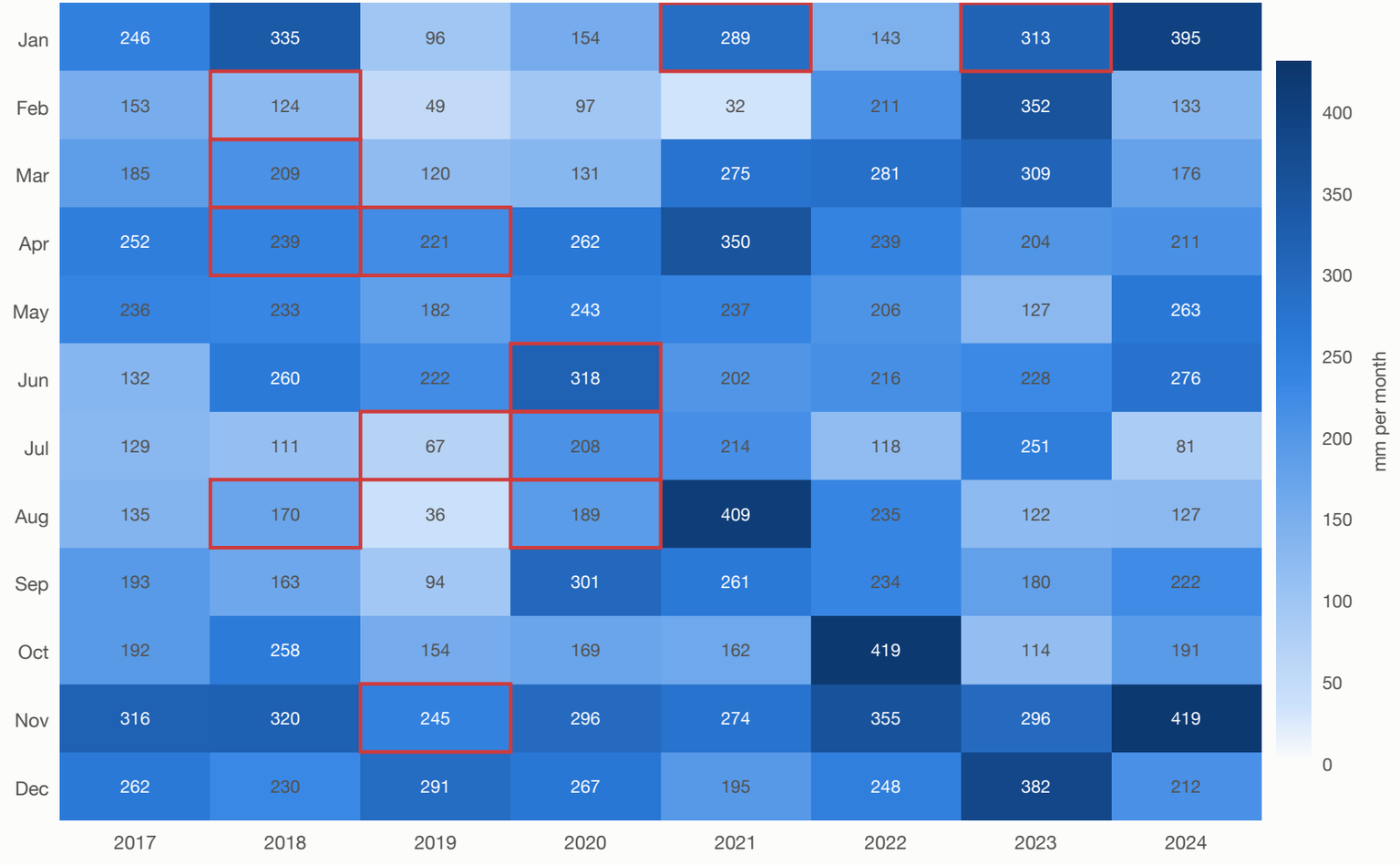
Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.

Wet-day counts are rescaled to a 365-day year so that years with missing days stay comparable.

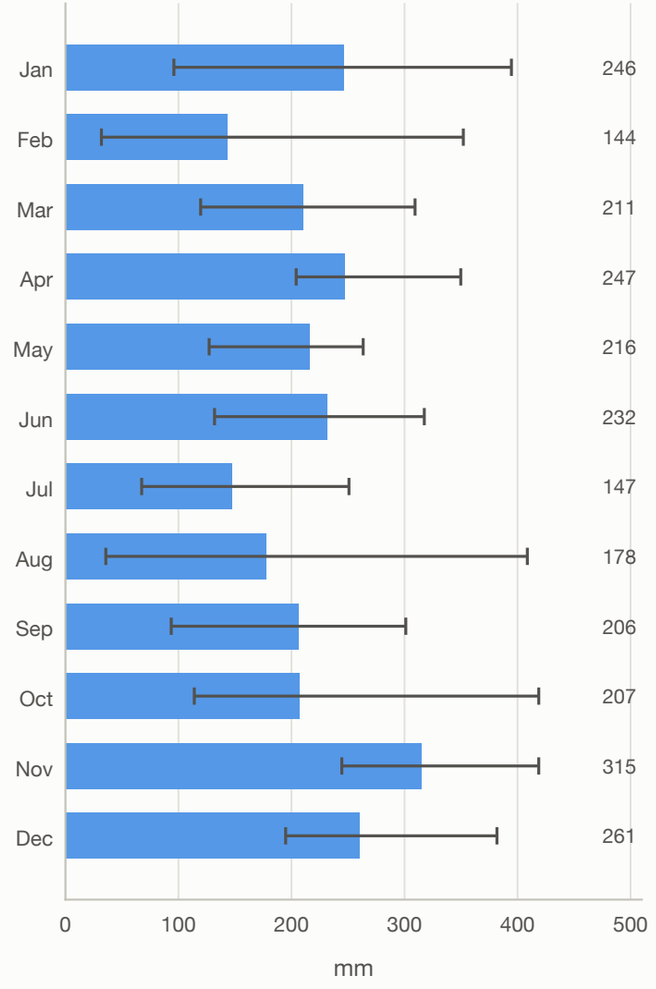
A double wet season, and a year-to-year spread that swamps it

November–January (north-east monsoon) and April–May (inter-monsoon) are the wet peaks; February and July are the dry troughs. But the same calendar month ranges over a factor of ten between years. February spans 32 mm (2021) to 352 mm (2023), Any monthly climatology for Singapore is therefore a weak predictor of any individual month.

a Monthly island rainfall, 2017–2024 (mm)



b Climatology (bars) and the 2017–2024 range (whiskers)

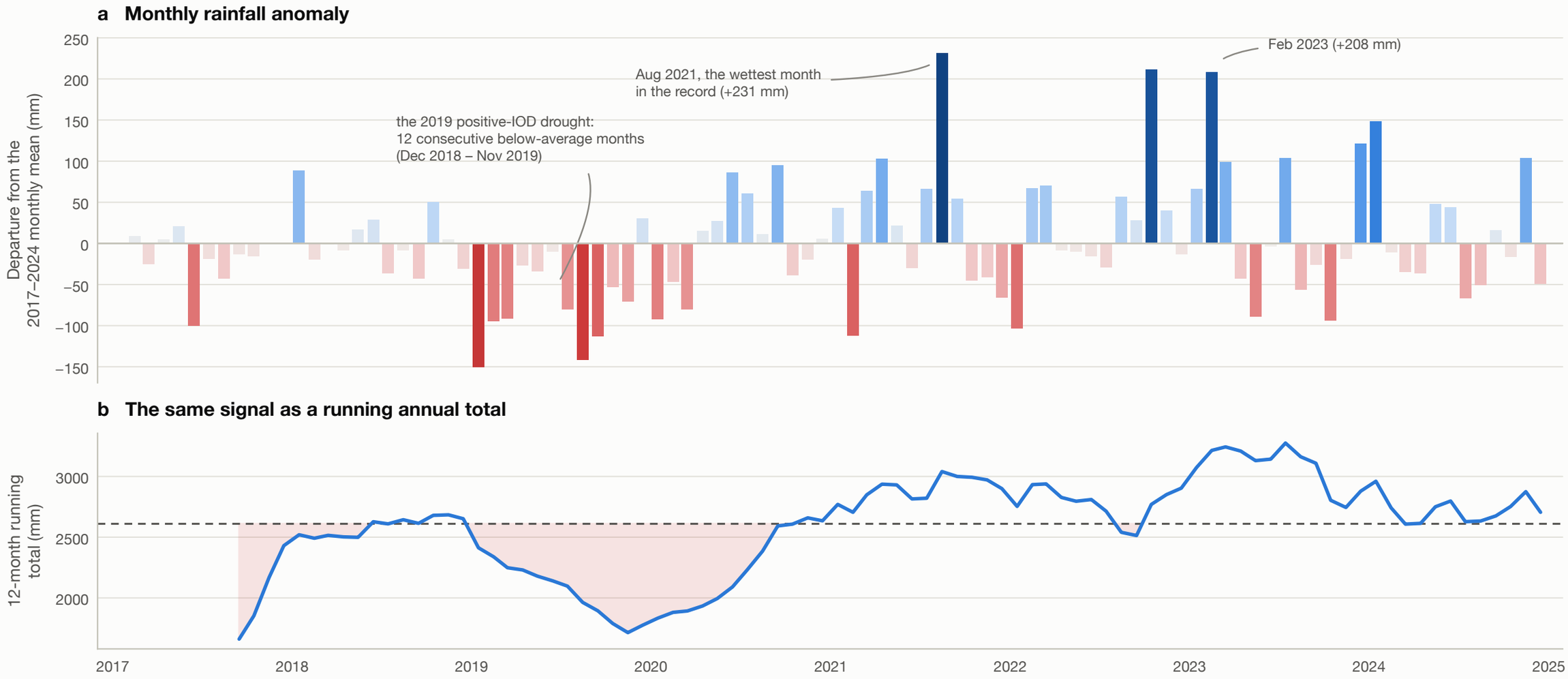


Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.

Red outline: month whose source record is <90% complete, filled from the day-of-year climatology of the other years.

Wet and dry spells in Singapore run for seasons, not for months

Anomalies are strongly autocorrelated: the 2019 drought runs 12 consecutive below-average months, and the 2021–2022 La Niña period delivers a sustained surplus. This persistence is what makes the year-to-year spread in Figure 3 so large. It is also a reason a short-horizon forecast (Task 2) should be given recent-rainfall state, not just a calendar position.



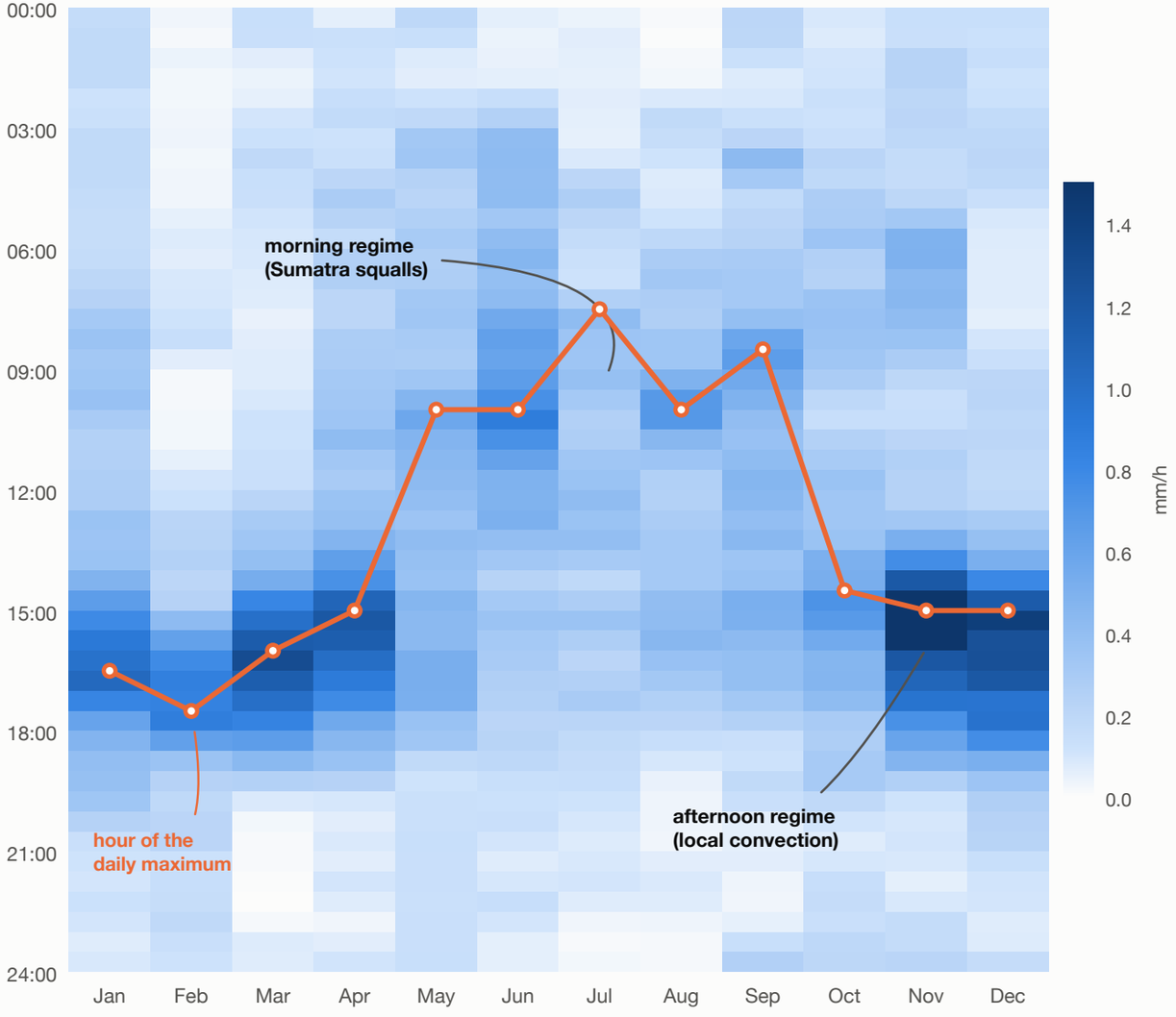
Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.

Anomalies are computed against each calendar month's own 2017–2024 mean, from the gap-filled island daily series.

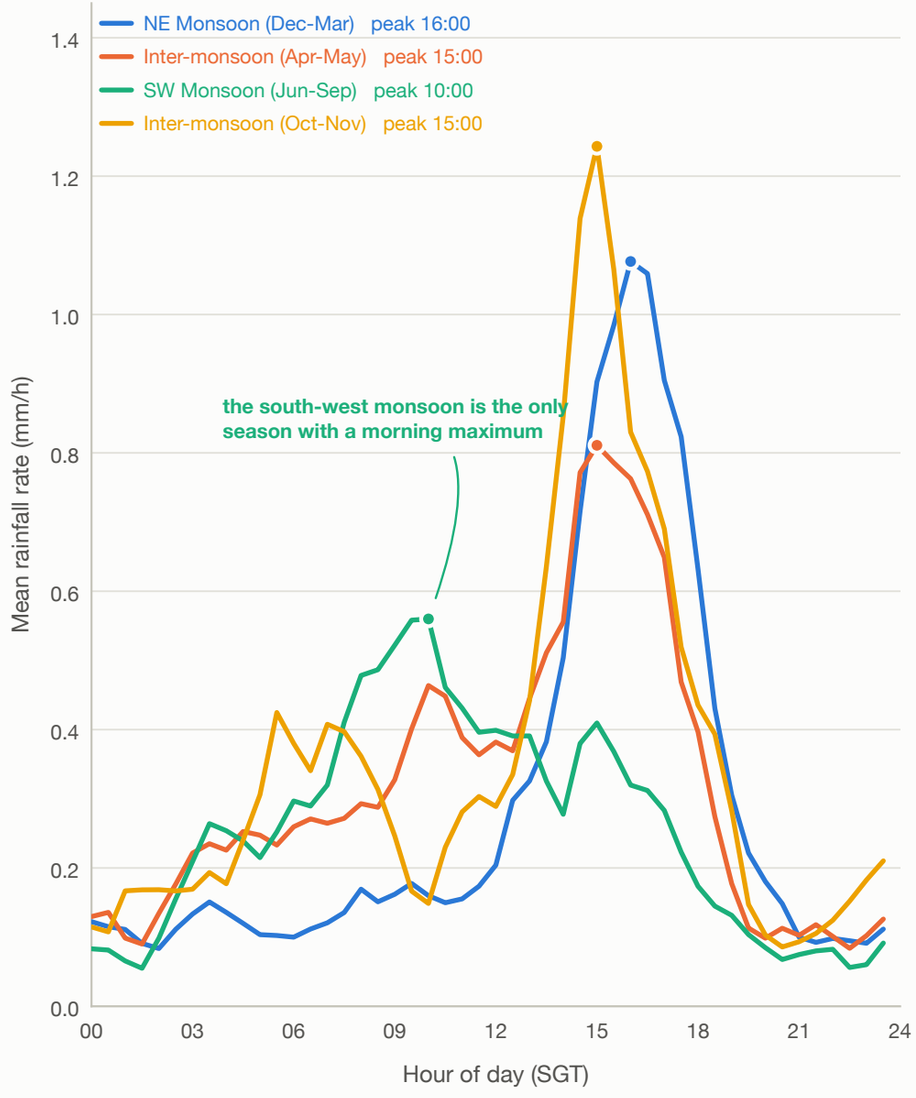
Singapore has two rainfall clocks, and the monsoon decides which one runs

From June to September the south-west monsoon shifts the daily rainfall maximum into the morning. This is the signature of Sumatra squall lines that form over the Malacca Strait overnight and sweep east across the island at dawn. In every other season the maximum is a sharp mid-afternoon convective peak near 15:00–16:00. The peak hour swings by roughly six hours between the two regimes.

a Mean rainfall rate by time of day and month (mm/h)



b The clock changes with the monsoon

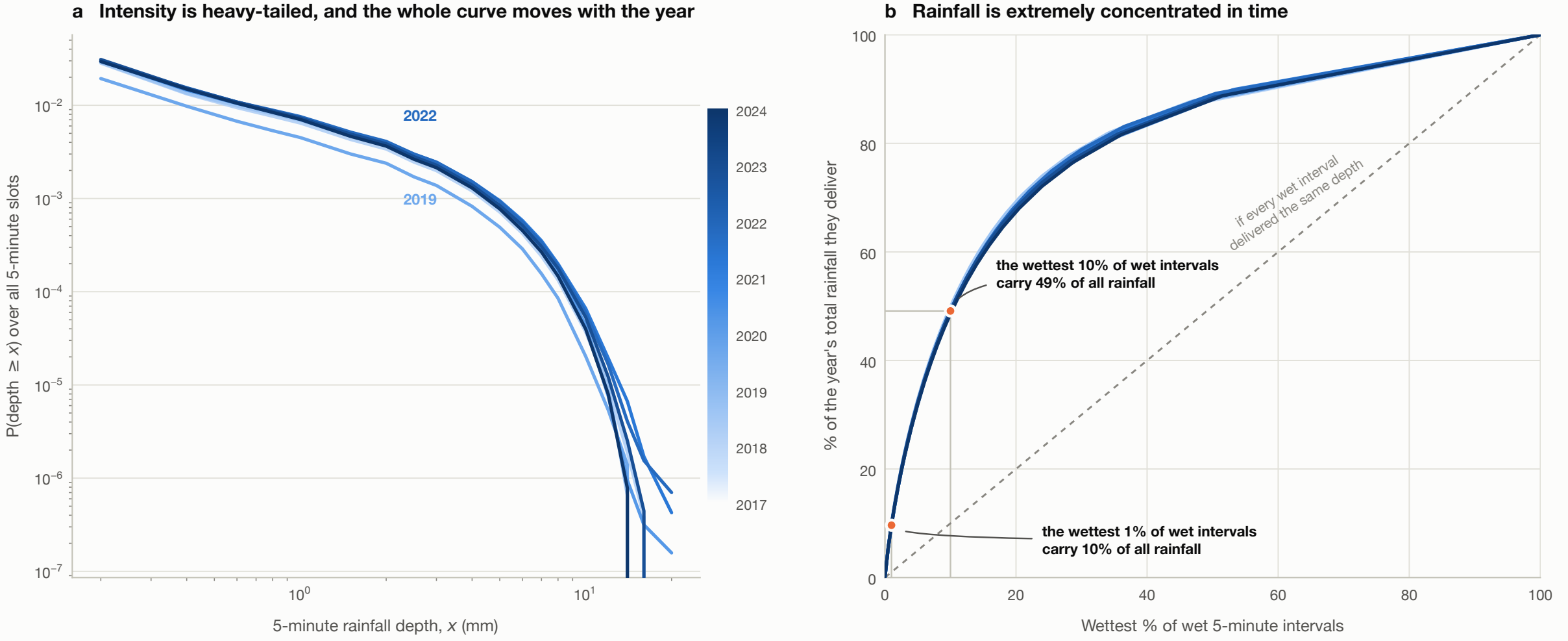


Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.

Rates are means over all stations and all 5-minute slots in each half-hour bin (dry slots included), so panel b is a true expected rate.

Almost half the rain that falls on Singapore arrives in 10% of the wet minutes

The concentration curve is near-identical in every year, including the drought: a dry year is not a year of gentler rain, it is a year with fewer rain events. The heavy tail in panel a is why a rainfall model should be framed as an exceedance probability rather than a conditional-mean regression, which is exactly how Task 2 is set up.

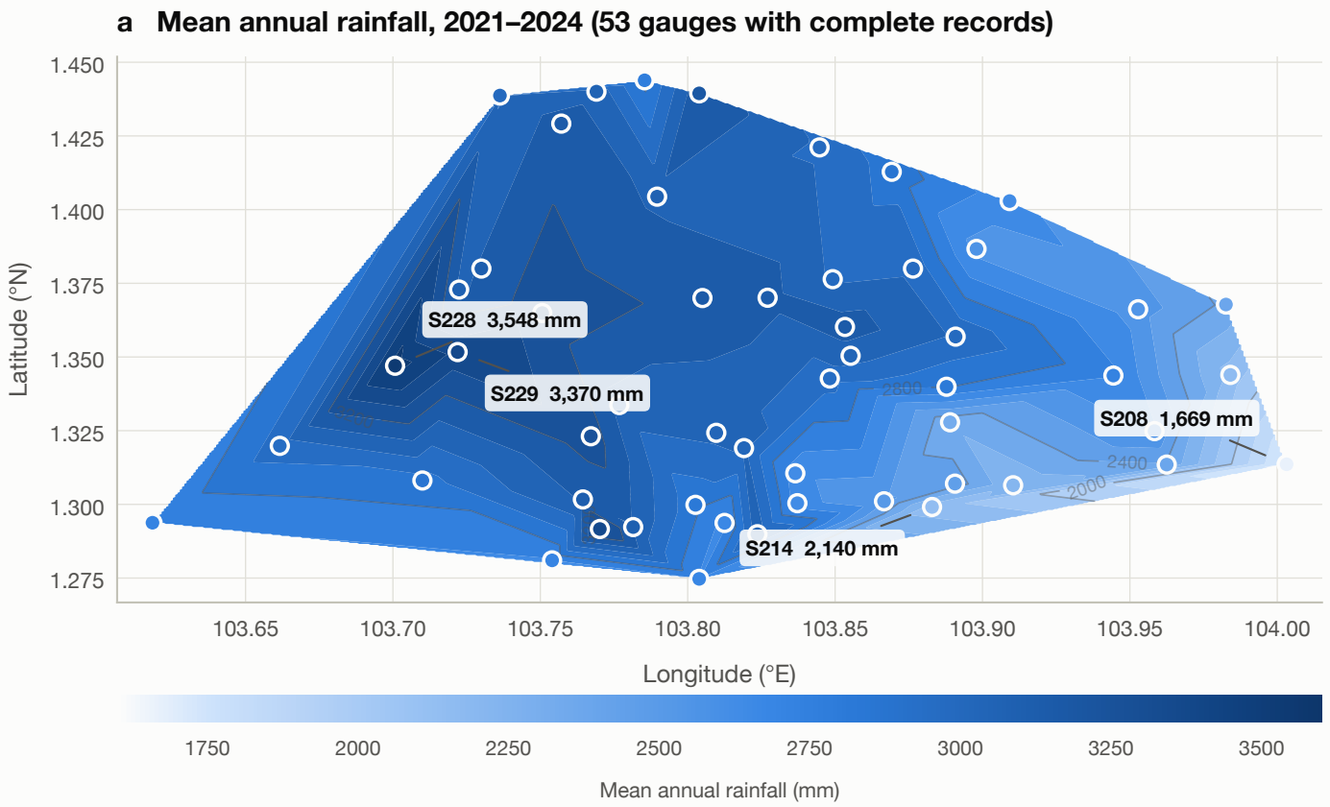


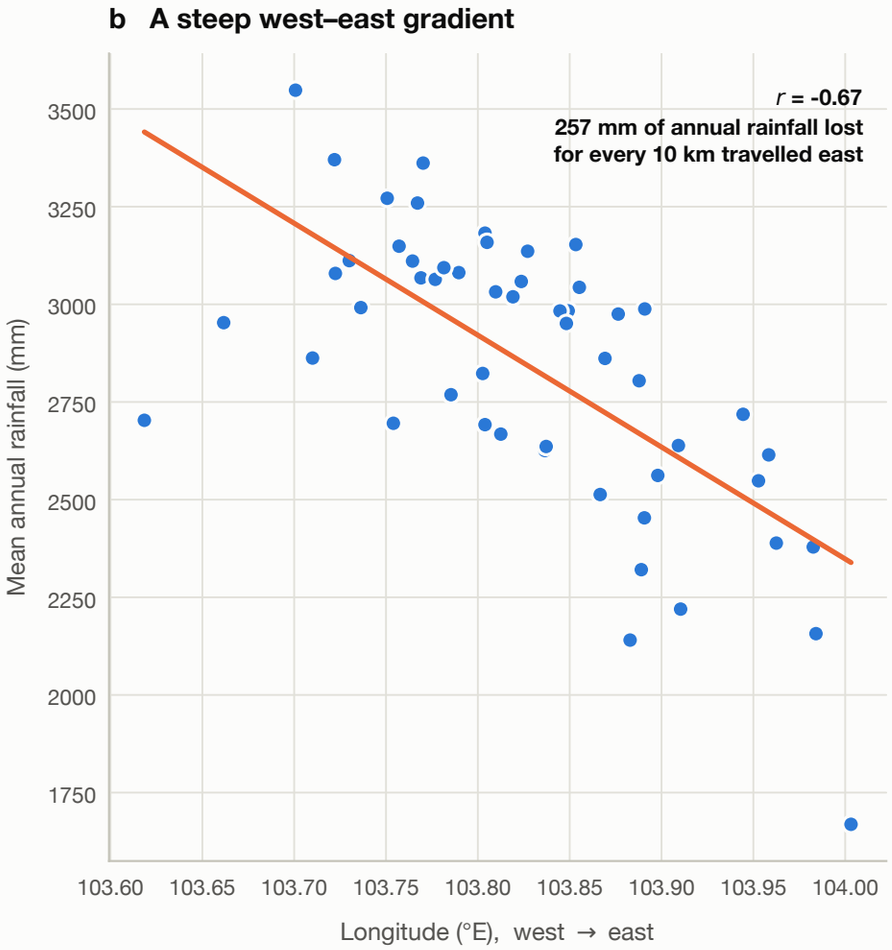
Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.

"Wet" = a 5-minute slot recording ≥0.2 mm (one tipping-bucket resolution step).

Rainfall more than doubles across an island 50 km wide

Over the common 2021–2024 period the wettest gauge records 3,548 mm a year and the driest 1,669 mm, a factor of 2.1.
The wet core sits over the western and central catchments, where sea-breeze convergence and higher terrain favour convection;
the east and the offshore islands are markedly drier. Station identity therefore carries real predictive information, and Task 2 keeps it as a feature.

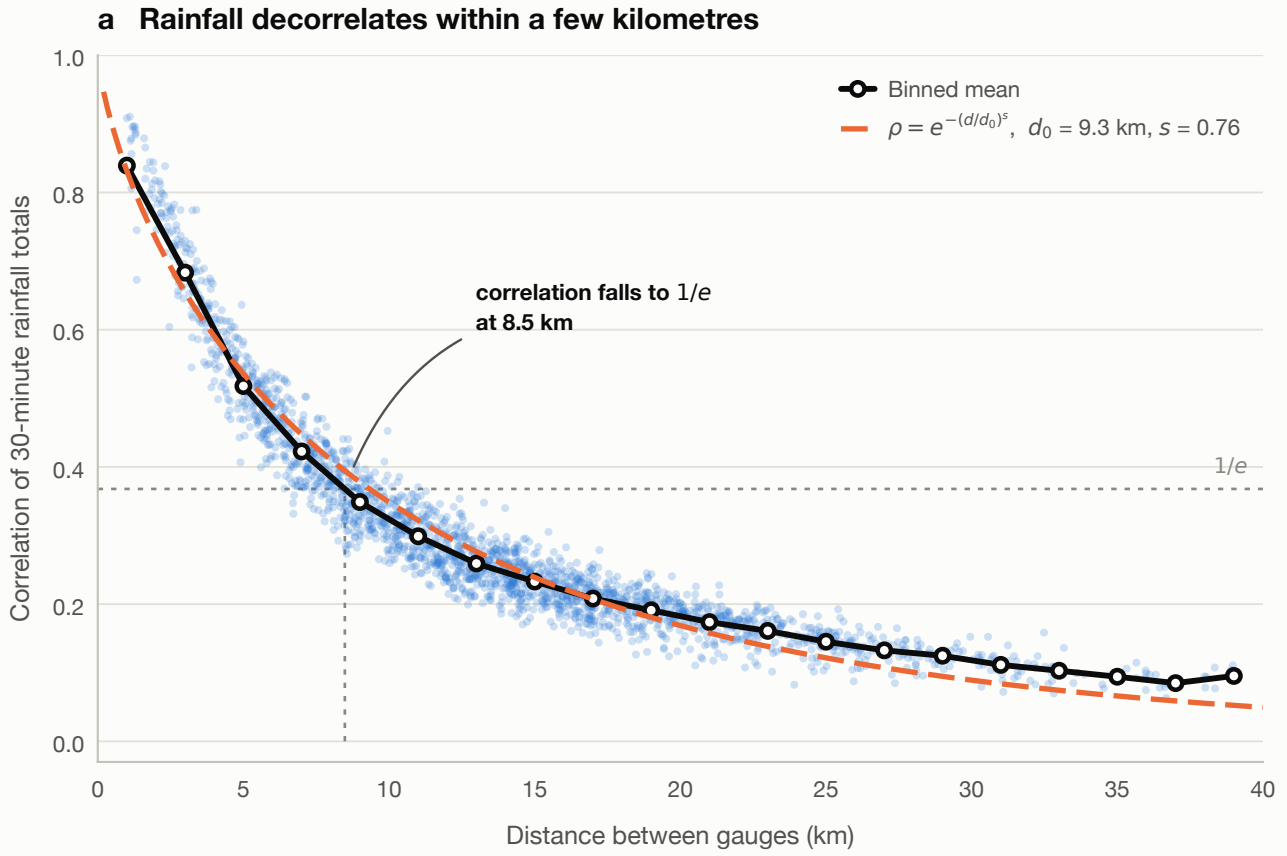


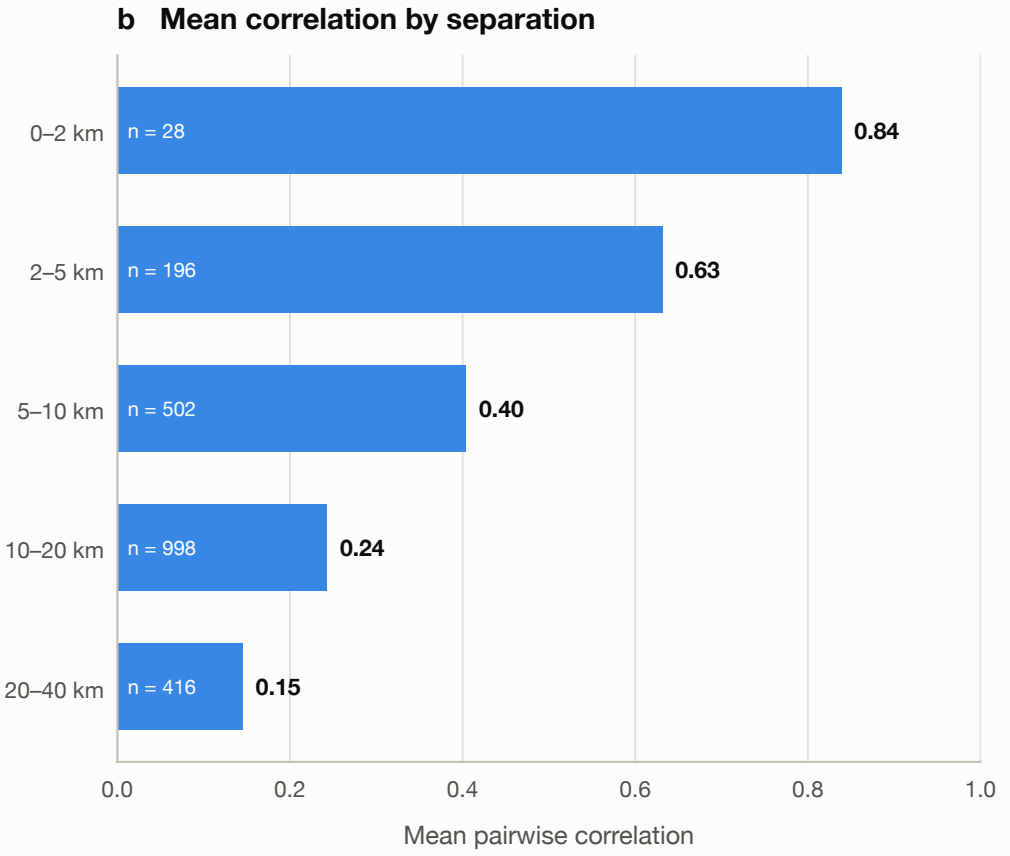


Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
A fixed four-year window is used so that gauges commissioned at different times are never compared over different periods. The field in panel a is a linear interpolation across the gauge network and is not extrapolated beyond the convex hull of the gauges.

Singapore rainfall loses coherence over about 8 km, a fraction of the island's width

Gauges less than 2 km apart agree closely on their 30-minute totals (ρ = 0.84); beyond 20 km they are nearly independent (ρ = 0.15).
That is the fingerprint of convective, cell-scale rainfall rather than broad frontal systems, and it carries a concrete modelling consequence:
a single island-wide forecast cannot be right everywhere at once, so Task 2 predicts per station and draws on neighbouring gauges only at short range.

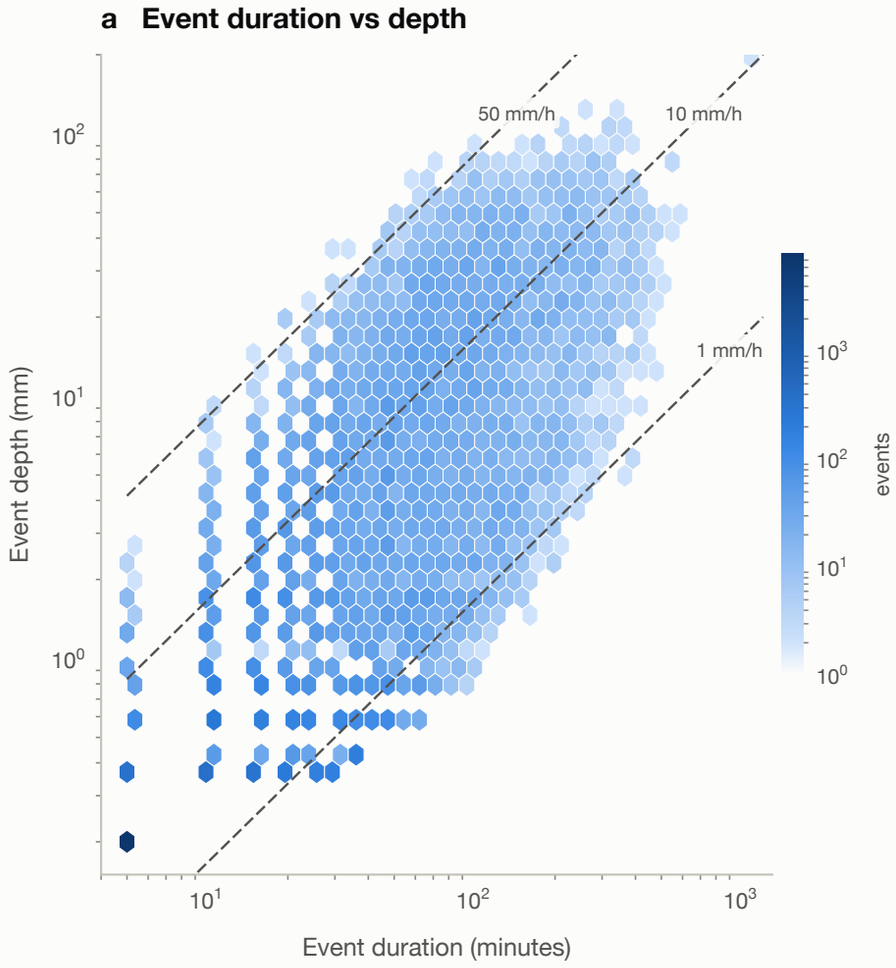


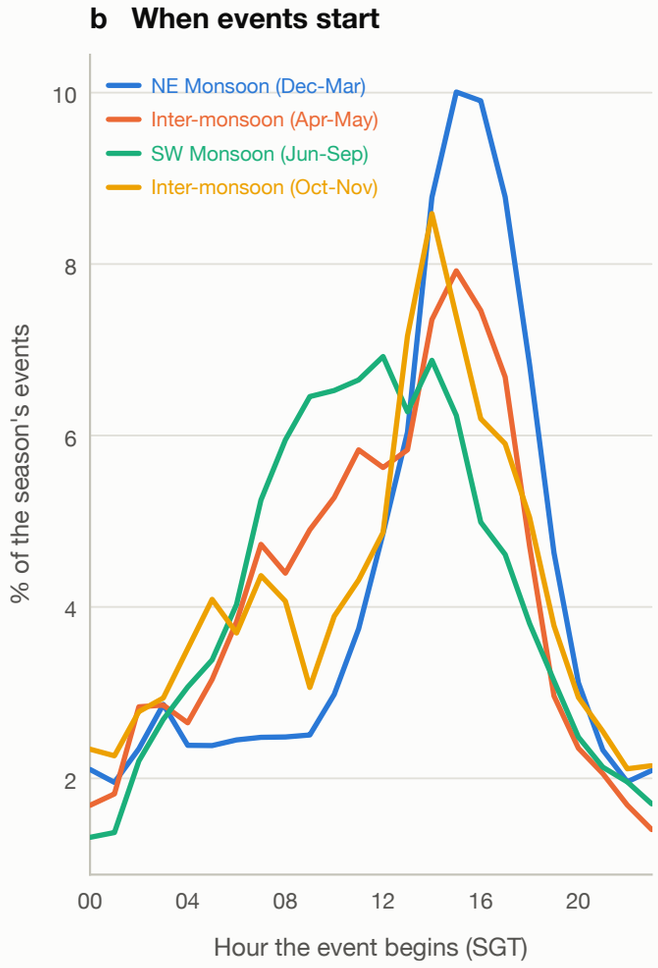


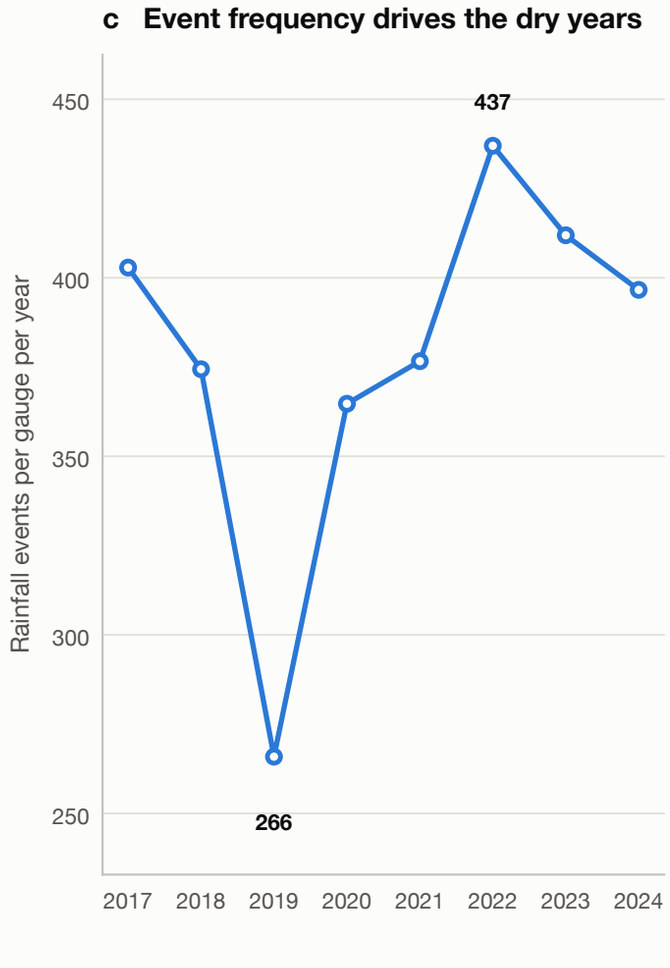
Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
2,145 station pairs, Pearson correlation of concurrent 30-minute totals over 2017–2024 (66 gauges with ≥4 complete years; pairwise-complete observations, ≥5,000 overlapping half-hours per pair). The 8.5 km figure is read off the binned means and assumes no functional form.

The typical rainfall event is short, light, and begins when the season says it should

Splitting each gauge's record into events separated by ≥30 minutes of dry weather gives a median duration of 20 minutes and a median depth near 1 mm, with a long tail of hour-scale storms along the 10–50 mm/h lines in panel a. Panel b recovers the same two regimes as Figure 5 from a completely different construction, and panel c shows the 2019 drought as a deficit of events, not of event size.





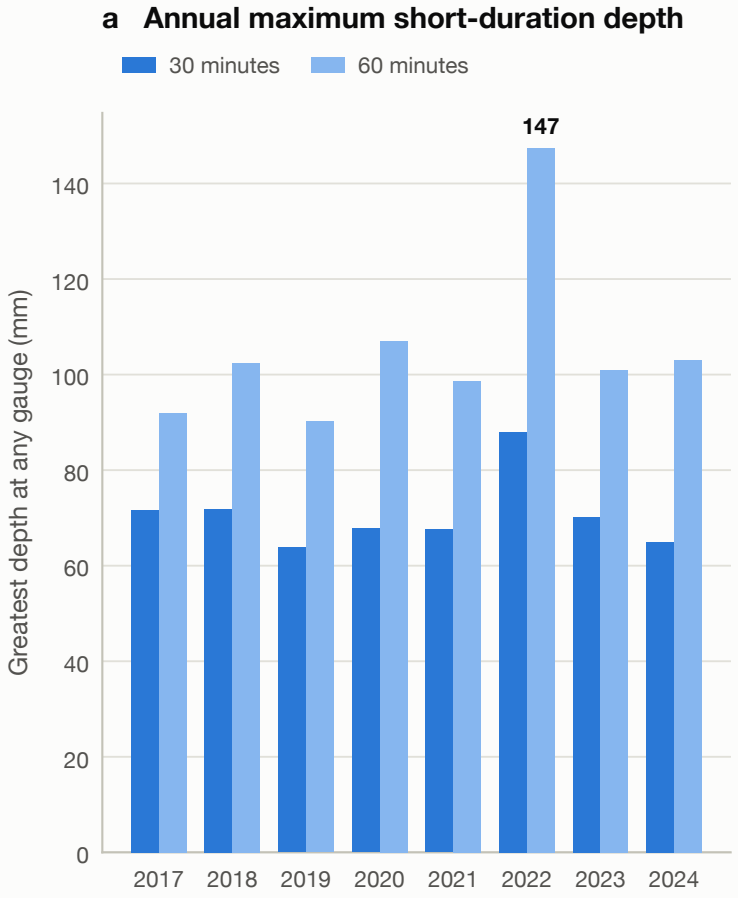


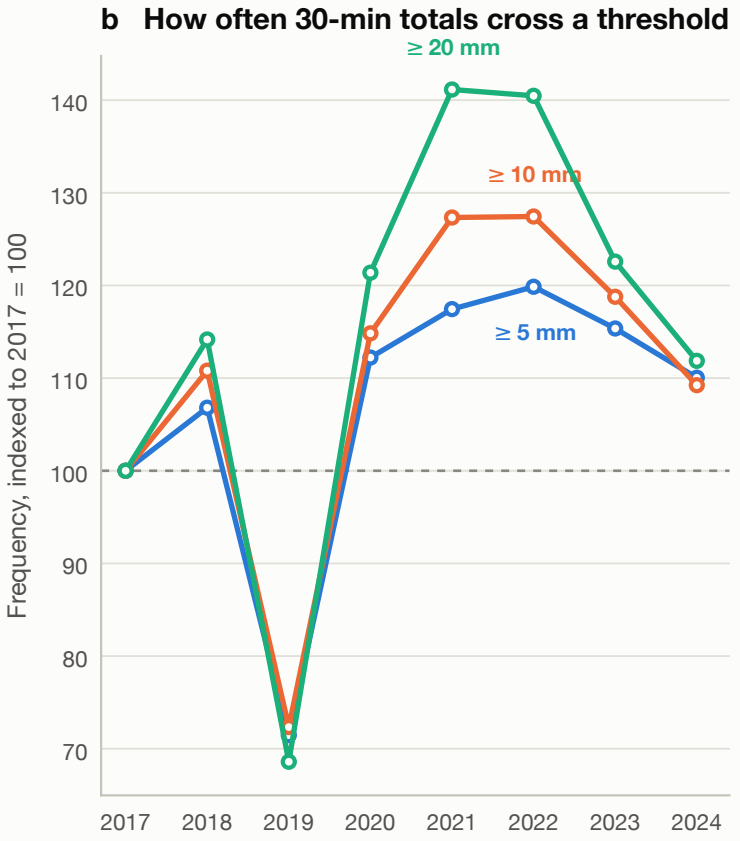
Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.

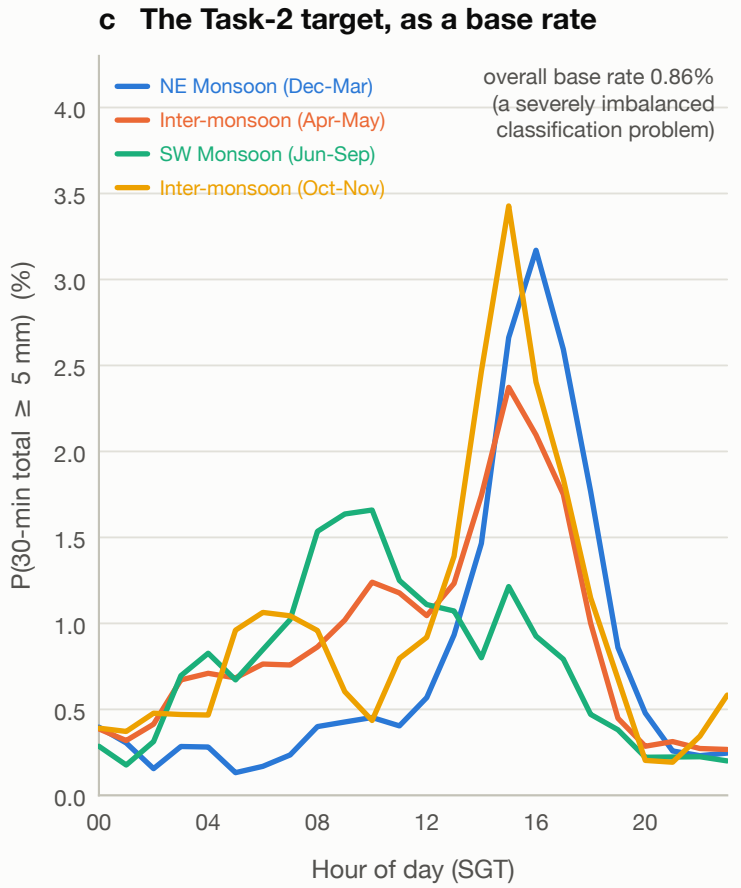
An event is a maximal run of wet 5-minute slots at one gauge with no dry gap exceeding 30 minutes. Panel a plots a 12% random sample for legibility.

Heavy half-hours became about 20% more frequent after 2020, and they keep the same daily rhythm

Panel b indexes the frequency of 5, 10 and 20 mm half-hours to 2017. All three rise together into the wet 2021–2022 period and stay elevated, so the shift is in how often heavy rain occurs rather than in how heavy the heaviest event gets. The annual maxima in panel a show no trend. Panel c is the quantity Task 2 must estimate, and it makes the class imbalance and the hour × season structure explicit.







Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.

30- and 60-minute accumulations are trailing rolling sums computed per gauge over the 5-minute series.